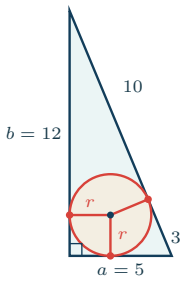
GEOMETRY ON ONE PAGE

Inscribed circles of Pythagorean triangles

A right triangle with whole sides has a whole inradius, so its incircle area is a square times π.

1 / THE RULE AND THE PICTURE

For a right triangle the inscribed circle has radius $r = \dfrac{a+b-c}{2}$



circle area $\pi r^2 = 4\pi$
hypotenuse $13 = 3 + 10$

The circle touches each side once, and the two tangent lengths from any vertex are equal. At the right angle the centre, the two touching points and the vertex form a square of side r, so the legs split as $a = r + (a - r)$ and $b = r + (b - r)$, and the hypotenuse is the sum of the two remaining tangents:

$$\boxed{c = (a - r) + (b - r) \implies r = \tfrac{a+b-c}{2}}$$

For the legs 5 and 12 the hypotenuse is 13, so $r = \frac{5+12-13}{2} = 2$ and the circle has area $\pi \cdot 2^2 = 4\pi \approx 12.57$ m^2, with no pen needed.

2 / THREE WAYS TO THE SAME RADIUS

Tangent lengths, area and the Euclid parameters agree

method	why it works	the 5, 12, 13 triangle	r
tangent lengths	tangents r, r at the right angle and $c = (a - r) + (b - r)$	$r = \frac{a+b-c}{2} = \frac{5+12-13}{2}$	2
area over half-perimeter	area $= rs$, so $r = \frac{ab}{a+b+c}$	$r = \frac{60}{30}$	2
Euclid parameters	$a+b-c = 2n(m-n)$ for $(m^2 - n^2, 2mn, m^2 + n^2)$	$m = 3,\ n = 2:\ r = 2 \cdot 1$	2

3 / THE INCIRCLE AREAS OF THE STANDARD TRIPLES

Always π times a perfect square

(m, n)	triple a, b, c	r	circle area
$(2, 1)$	$3, 4, 5$	1	1π
$(3, 2)$	$5, 12, 13$	2	4π
$(4, 1)$	$15, 8, 17$	3	9π
$(4, 3)$	$7, 24, 25$	3	9π
$(5, 2)$	$21, 20, 29$	6	36π

The radius is a whole number for every Pythagorean triple, because $a+b+c$ is always even and so $a+b-c$ is too. With the Euclid parameters it is $n(m-n)$, and for $5, 12, 13$ the circle fills $\frac{4\pi}{30} = 0.419$ of the triangle.

4 / RUNNING IT BACKWARDS

Every whole right triangle with inradius r comes from a factor pair of $2r^2$

Putting $c = a + b - 2r$ into $a^2 + b^2 = c^2$ gives $ab - 2r(a + b) + 2r^2 = 0$, which factorises as

$$\boxed{(a - 2r)(b - 2r) = 2r^2}$$

so each factor pair $u \cdot v = 2r^2$ gives a triangle with legs $2r + u$ and $2r + v$. For $r = 2$ we have $2r^2 = 8 = 1 \cdot 8 = 2 \cdot 4$, giving $5, 12, 13$ and $6, 8, 10$, so the same area 4π belongs to two different triangles, and in general the number of triangles is half the number of divisors of $2r^2$.

r	triangles	which triangles (legs $a \le b$, hypotenuse c)
2	2	$5, 12, 13$ and $6, 8, 10$
3	3	$7, 24, 25$ and $8, 15, 17$ and $9, 12, 15$
6	6	$13, 84, 85$; $14, 48, 50$; $15, 36, 39$; $16, 30, 34$; $18, 24, 30$; $20, 21, 29$

5 / THE CIRCUMCIRCLE AND EULER'S RELATION

The hypotenuse is a diameter of the outer circle

The circumcircle has radius $R = \frac{c}{2} = 6.5$, so the radii are in the ratio $\frac{r}{R} = \frac{4}{13}$ and the circle areas in the ratio $\frac{16}{169}$. Euler's relation $OI^2 = R(R - 2r) = 16.25$ matches the centres $(2, 2)$ and $(2.5, 6)$.

6 / TRY THESE

Radius first, then area

(a) Legs 8 and 15 give hypotenuse 17, so $r = \frac{8+15-17}{2} = 3$ and the area is 9π. **(b)** Legs 20 and 21 give $r = 6$ and area 36π. **(c)** Every triangle with area 9π has $2r^2 = 18 = 1 \cdot 18 = 2 \cdot 9 = 3 \cdot 6$, giving $7, 24, 25$, $8, 15, 17$ and $9, 12, 15$.

ASIDE / ANY TRIANGLE

For any triangle $r = \text{area}/s$, as in the second method above.

LINKS TO OTHER SHEETS

The Euclid parameters are on the Pythagorean triples sheets; factor pairs also drive the difference of two squares.

Convention. a, b legs; c hypotenuse; r inradius; R circumradius; s semi-perimeter; $m > n$ coprime, opposite parity.

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